

TEST OUTPUT FREQUENCIES FOR IC1 OP-AMP WITH SYMMETRICAL POWER SUPPLY, (VCC+, VCC- AND GND) AT $\pm 5V$

When switch is closed	Selected capacitor	Value of the capacitor	Approximate test output frequency
S1	C1	2.2 μ F	30Hz
S2	C2	220nF	300Hz
S3	C3	22nF	3kHz
S4	C4	2.2nF	33kHz
None	C5	220pF	330kHz

Note: Frequencies may vary depending on op-amps and capacitors' tolerances.

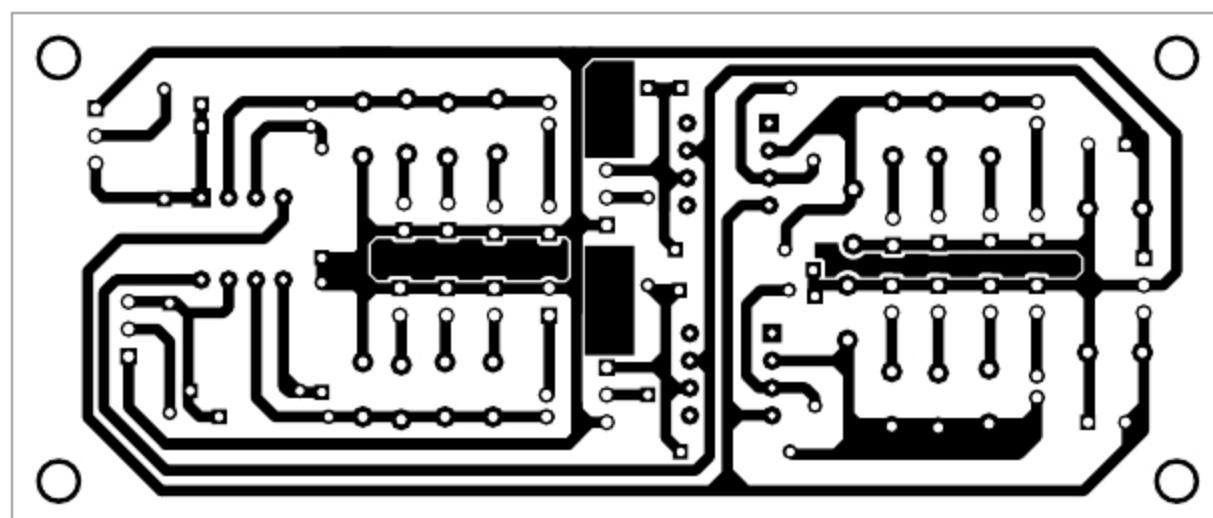


Fig. 2: Actual-size PCB layout of the simple op-amp tester

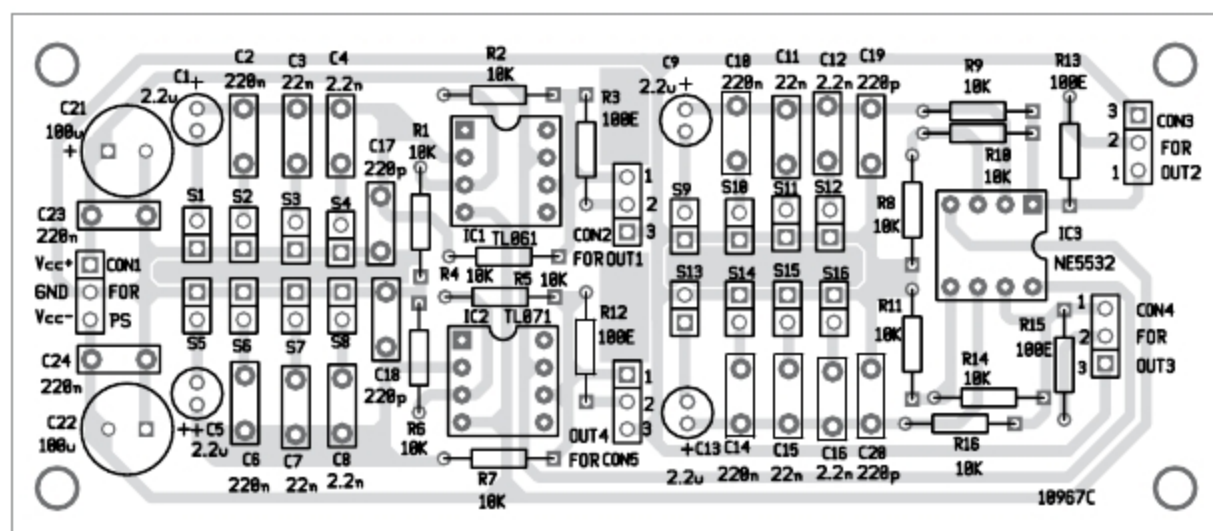


Fig. 3: Components layout for the PCB

well as output with series resistor of 100 ohms. Op-amps will be unstable if you test them directly with capacitive and inductive loads, so 100-ohm resis-

tors are provided at output.

The circuit does not need any adjustment to operate properly. It can be used to test many op-amps, includ-

PARTS LIST

Semiconductors:

- IC1 - TL061 op-amp
- IC2 - TL071 op-amp
- IC3 - NE5532 op-amp

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R4-R11, R14, R16 - 10-kilo-ohm
- R3, R12, R13, R15 - 100-ohm

Capacitors:

- C1, C5, C9, C13 - 2.2 μ F, 35V electrolytic
- C2, C6, C10, C14, C23, C24 - 220nF ceramic disk
- C3, C7, C11, C15 - 22nF ceramic disk
- C4, C8, C12, C16 - 2.2nF ceramic disk
- C17-C20 - 220pF ceramic disk
- C21, C22 - 100 μ F, 35V electrolytic

Miscellaneous:

- CON1-CON5 - 3-pin connector
- S1-S16 - On/off switch
- PS - ± 5 to ± 15 dual power supply
- 8-pin base (3 No.)

ing low-speed op-amps such as LM741 and MC1458, and faster op-amps such as OPA132 and OPA134.

Construction and testing

An actual-size PCB layout of the simple tester for single and dual op-amps is shown in Fig. 2 and its components layout in Fig. 3. Assemble the circuit on the designed PCB. Connect a suitable $\pm$ power supply (PS) across CON1. Connect all switches (S1 through S16) on the front of the cabinet to select a particular frequency, as shown in the table.

It is recommended to use IC bases for IC1, IC2 and IC3, so that you can insert and remove them easily. **EFY**



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